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EXPERIMENT 4

Implementation of K-Nearest Neighbors (KNN) and Performance Evaluation

K-Nearest Neighbors (KNN) Algorithm

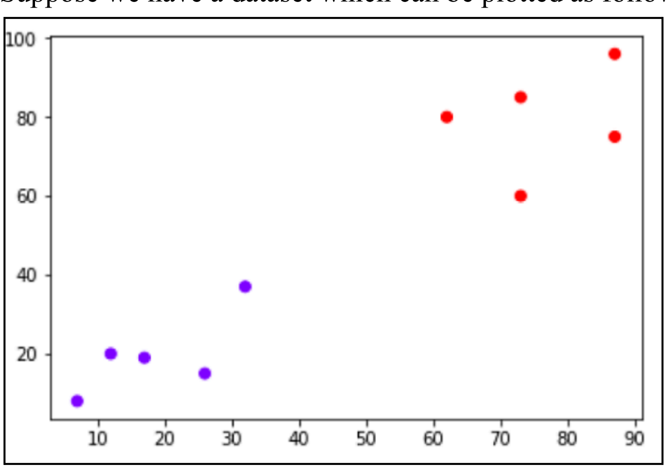
K-nearest neighbors (KNN) algorithm is a type of supervised ML algorithm which can be used for both classification as well as regression predictive problems. However, it is mainly used for classification of predictive problems in industry. The main idea behind KNN is to find the k-nearest data points to a given test data point and use these nearest neighbors to make a prediction. The value of k is a hyperparameter that needs to be tuned, and it represents the number of neighbors to consider.

For classification problems, the KNN algorithm assigns the test data point to the class that appears most frequently among the k-nearest neighbors. In other words, the class with the highest number of neighbors is the predicted class.
For regression problems, the KNN algorithm assigns the test data point the average of the k-nearest neighbors' values.

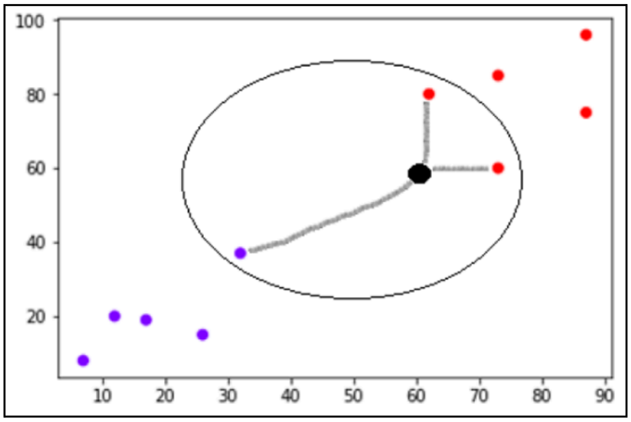
The distance metric used to measure the similarity between two data points is an essential factor that affects the KNN algorithm's performance. The most commonly used distance metrics are Euclidean distance, Manhattan distance, and Minkowski distance.

Example

The following is an example to understand the concept of K and working of KNN algorithm −
Suppose we have a dataset which can be plotted as follows −



Now, we need to classify a new data point with black dot (at point 60,60) into a blue or red class. We are assuming K = 3 i.e. it would find three nearest data points. It is shown in the next diagram −



We can see in the above diagram the three nearest neighbors of the data point with black dot. Among those three, two of them lie in Red class hence the black dot will also be assigned in red class.

1.<u>Dataset Source</u>

Dataset Name: Heart Disease Prediction Dataset
Source: Kaggle
Link:
https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

File Used: `heart.csv`

This dataset is different from your previous experiments.

2.<u>Dataset Description</u>

This dataset predicts whether a person has heart disease.

📊 Dataset Info

- Total Samples: 303
- Features: 13
- Target: `target`
    - 0 → No Heart Disease
    - 1 → Heart Disease

Important Features :

Feature	Description
age	Age
sex	Gender
cp	Chest pain type
trestbps	Resting blood pressure
chol	Cholesterol
thalach	Max heart rate
oldpeak	ST depression
slope	Slope of ST segment
ca	Number of major vessels
thal	Thalassemia

Steps:

1. Choose K
2. Compute distance from test point to all training points
3. Select K nearest neighbors
4. Majority voting
5. Assign class

3.Mathematical Formulation of KNN

Distance Metrics Used in KNN Algorithm

KNN uses distance metrics to identify nearest neighbor, these neighbors are used for classification and regression task. To identify nearest neighbor we use below distance metrics:

1. Euclidean Distance

Euclidean distance is defined as the straight-line distance between two points in a plane or space. You can think of it like the shortest path you would walk if you were to go directly from one point to another.

$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^{d} (x_j - X_{i_j})^2}]$$

2. Manhattan Distance

This is the total distance you would travel if you could only move along horizontal and vertical lines like a grid or city streets. It's also called "taxicab distance" because a taxi can only drive along the grid-like streets of a city.

$$d\left(x, y\right) = \sum_{i=1}^{n} \left|x_i - y_i\right|$$

3. Minkowski Distance

Minkowski distance is like a family of distances, which includes both Euclidean and Manhattan distances as special cases.

$$d\left(x, y\right) = \left(\sum_{i=1}^{n} \left(x_i - y_i\right)^p\right)^{\frac{1}{p}}$$

From the formula above, when p=2, it becomes the same as the Euclidean distance formula and when p=1, it turns into the Manhattan distance formula. Minkowski distance is essentially a flexible formula that can represent either Euclidean or Manhattan distance depending on the value of p.

4.Algorithm Limitations

1.Slow for large datasets
2.Sensitive to scaling
3.Sensitive to irrelevant features
4.Choosing optimal K is difficult

5.Methodology / Workflow

Load Data → Preprocess → Scale → Train-Test Split → Train KNN → Evaluate → Tune K → Compare

.6.<u>Performance Analysis</u>

We evaluate using:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-score

Interpretation

- High Accuracy → Good classification
- High Recall → Detecting disease patients properly
- Balanced Precision & Recall → Stable model

7.<u>Hyperparameter Tuning</u>

Important KNN hyperparameters:

Parameter	Meaning
n_neighbors	Value of K
weights	uniform or distance
metric	Distance type

Effect of K

Small K:

- Overfitting
- Sensitive to noise

Large K:

- Underfitting
- Smoother decision boundary